UNIT 5

APPLICATION LAYER

- ✓ Domain name system
- ✓ SNMP
- ✓ Electronic mail
- ✓ The world WEB
- ✓ HTTP
- ✓ Streaming of Audio and Video

INTRODUCTION:

- The application layer in the OSI model is the closest layer to the end user which means that the application layer and end user can interact directly with the software application. **The application layer programs are based on client and servers.**

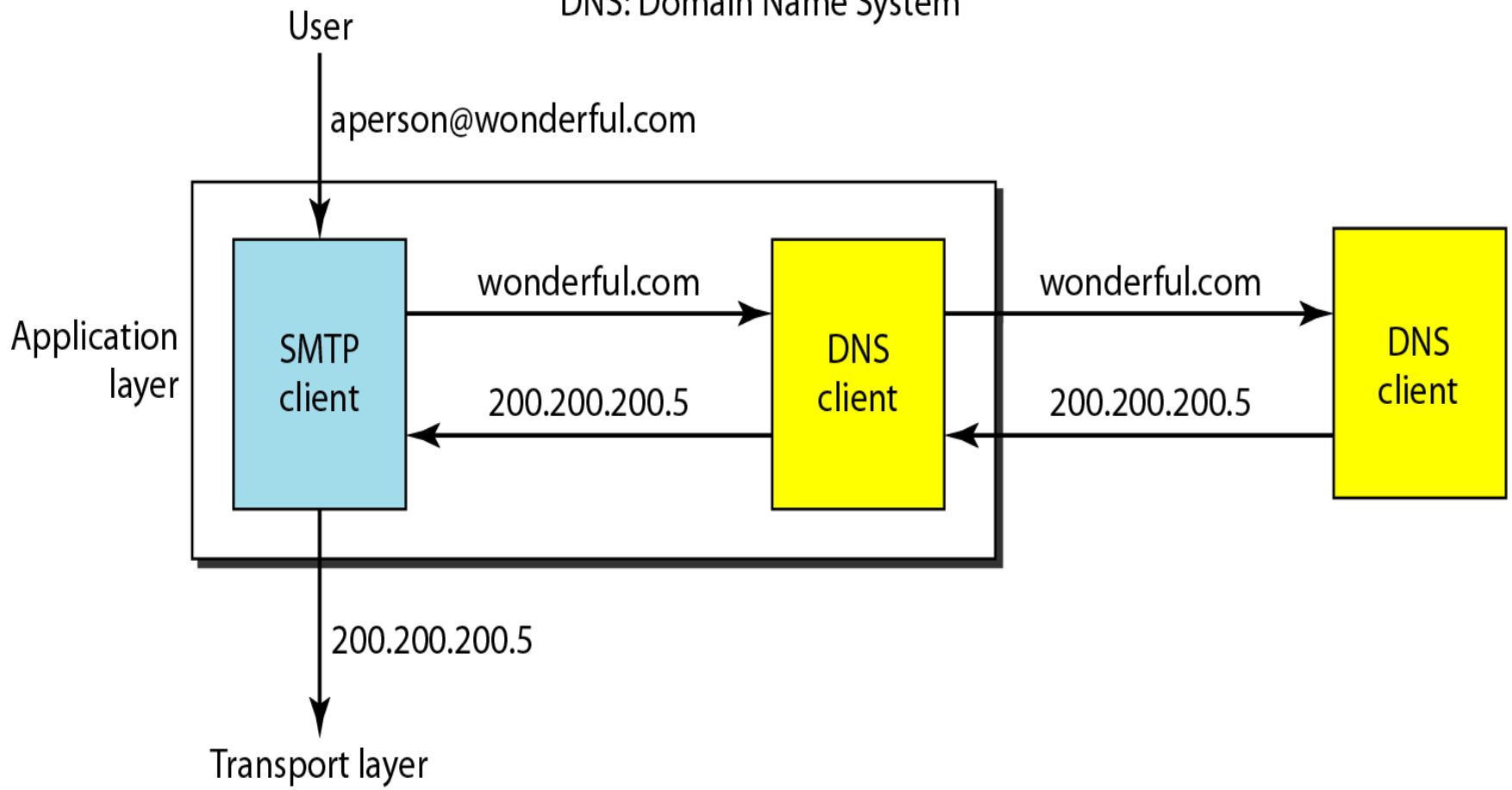
Functions:

- ✓ Identifying communication partners
- ✓ Determining resource availability
- ✓ Synchronizing communication

Domain Name System(DNS)

- ✓ An application layer protocol defines how the application processes running on different systems, pass the messages to each other.
- ✓ **DNS is a directory service that translates the domain name into IP addresses or an address to a name**. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- ✓ For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

SMTP: Simple Mail Transfer Protocol (e-mail)
DNS: Domain Name System



DOMAIN NAME SPACE:

A name space that maps each address to a unique name can be organized in two ways: flat or hierarchical.

Flat Name Space

In a flat name space, a name is assigned to an address. A name in this space is a sequence of characters without structure.

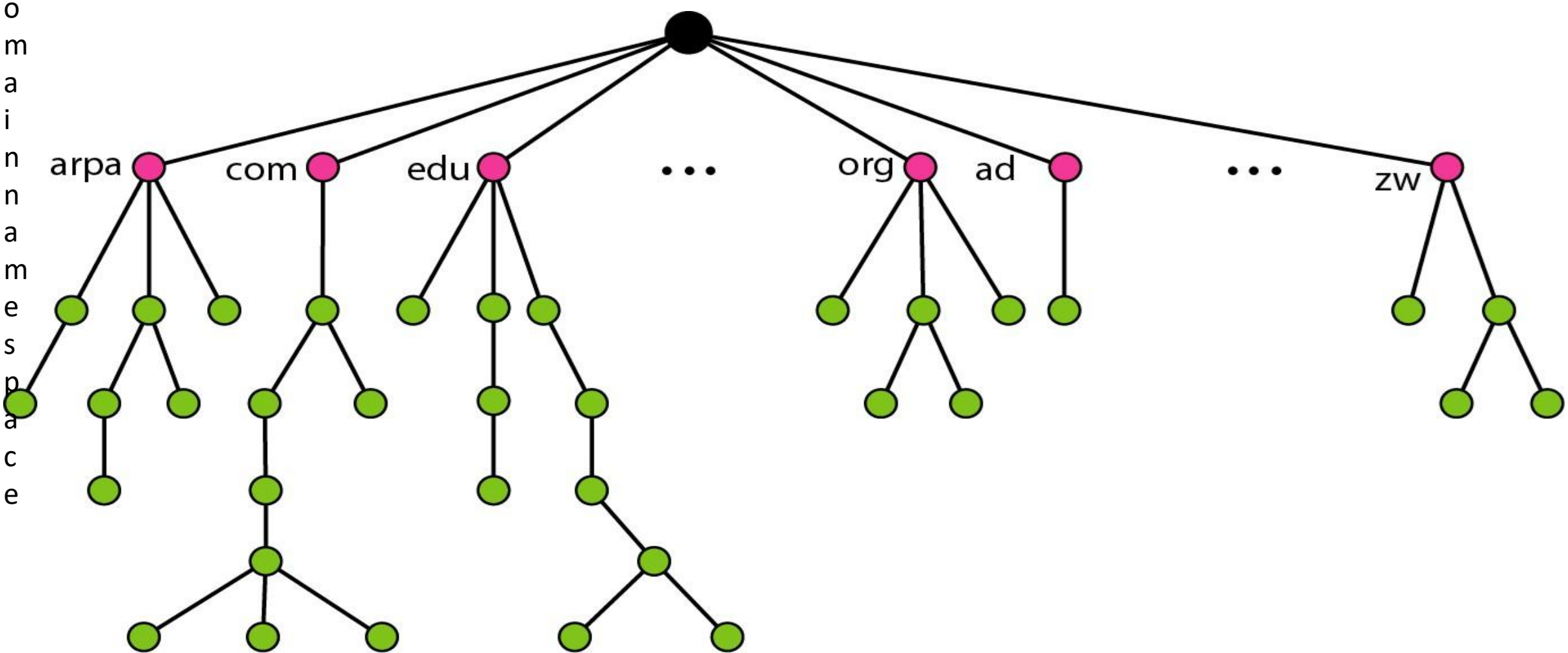
Hierarchical Name Space

In a hierarchical name space, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on.

Exa: *challenger.jhda.edu, challenger.berkeley.edu, and challenger.smart.com*

- To have a hierarchical name space, a domain name space was designed. **In this design the names are defined in an inverted-tree structure** with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127.

Domain name space

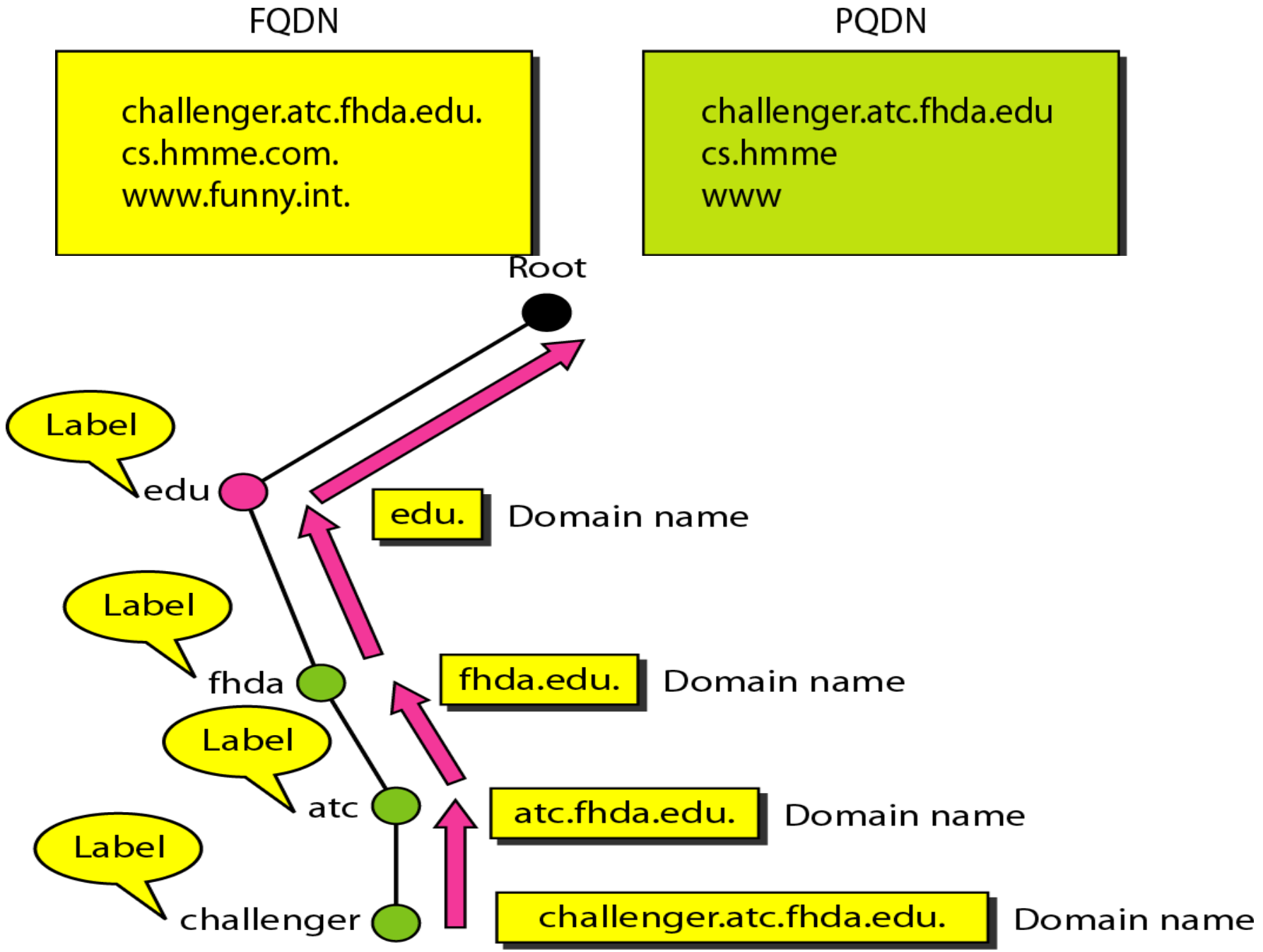


Label

Each node in the tree has a label, which is a string with a maximum of 63 characters. The root label is a null string (empty string). DNS requires that children of a node (nodes that branch from the same node) have different labels, which guarantees the uniqueness of the domain names.

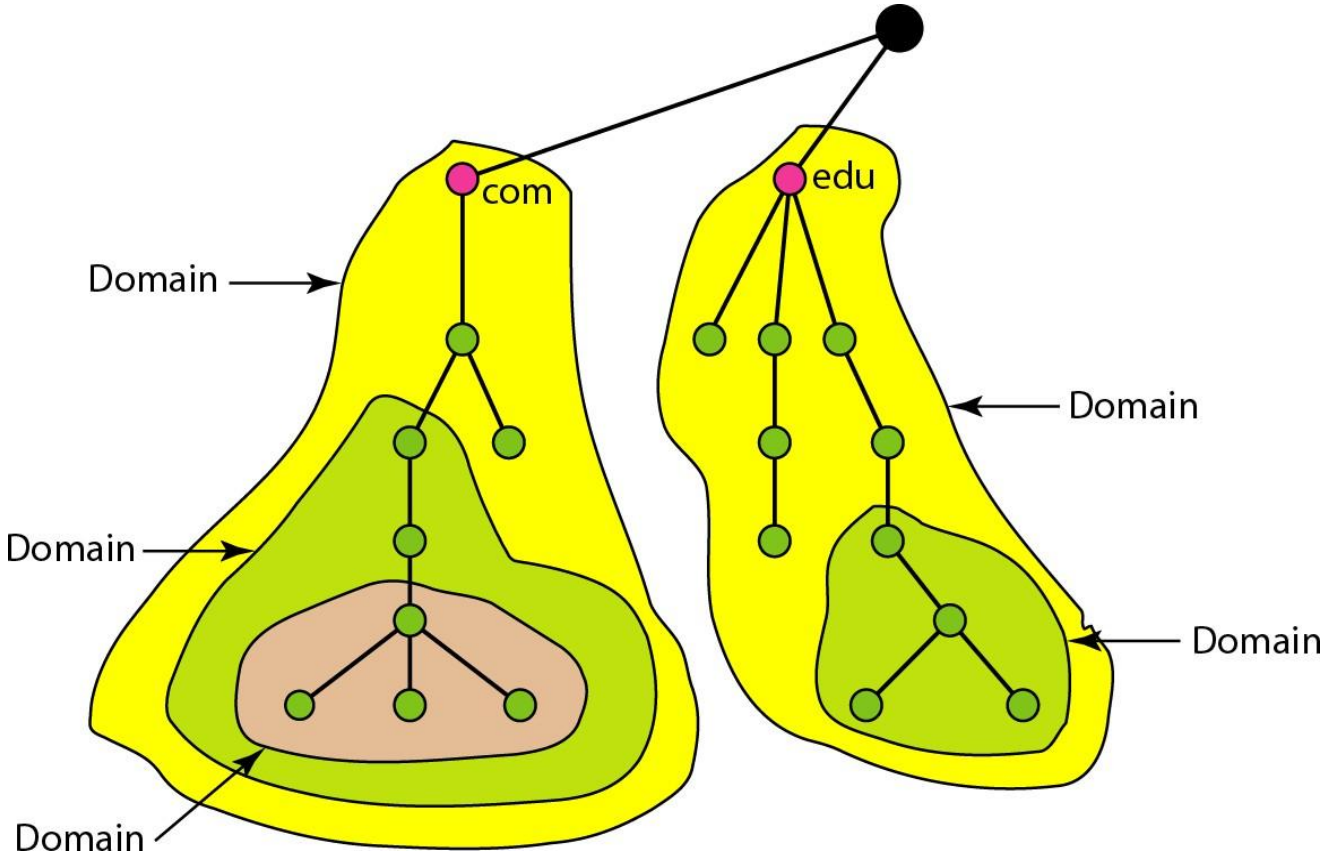
Domain Name

Each node in the tree has a domain name. A full domain name is a sequence of labels separated by dots (.). The domain names are always read from the node up to the root. The last label is the label of the root (null). This means that a full domain name always ends in a null label, which means the last character is a dot because the null string is nothing. Below Figure shows some domain names


FQDN
challenger.atc.fhda.edu.
cs.hmme.com.
www.funny.int.
PQDN
challenger.atc.fhda.edu
cs.hmme
www
Root
Label
edu
edu.
Domain name
Label
fhda
fhda.edu.
Domain name
Label
atc
atc.fhda.edu.
Domain name
Label
challenger
challenger.atc.fhda.edu.
Domain name

Domain

- A domain is a subtree of the domain name space. The name of the domain is the domain name of the node at the top of the subtree.

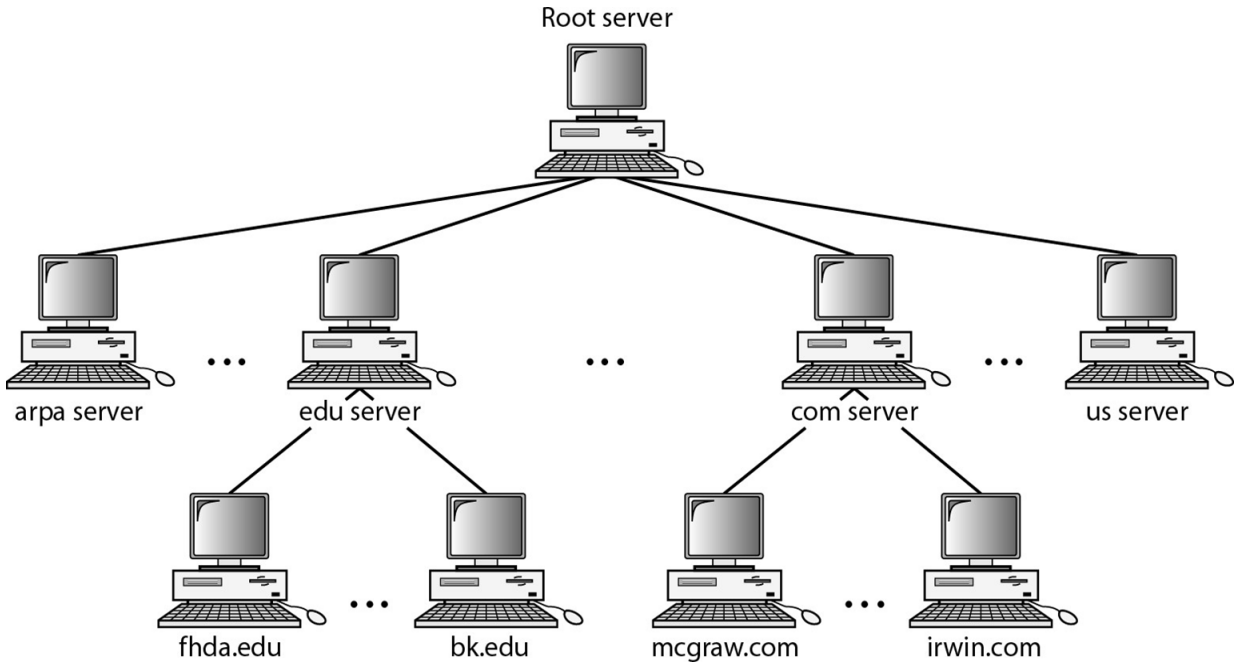


DISTRIBUTION OF NAME SPACE:

- The information contained in the domain name space must be stored. However, it is very inefficient and also unreliable to have just one computer store such a huge amount of information. In this section, we discuss the distribution of the domain name space.

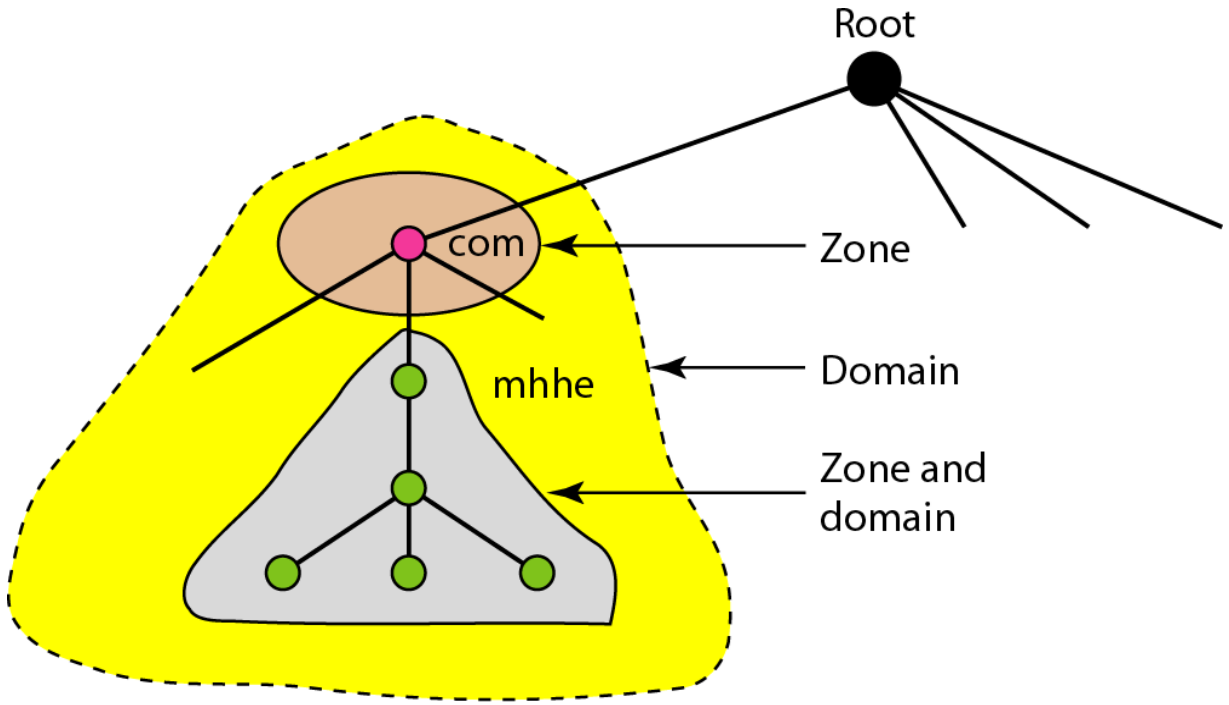
1. <u>Hierarchy of Name Servers</u>

➢ distribute the information among many computers called DNS servers. we let the root stand alone and create as many domains (subtrees) as there are first-level nodes.



2. Zone

- Since the complete domain name hierarchy cannot be stored on a single server, it is divided among many servers. What a server is responsible for or has authority over is called a zone. We can define a zone as a contiguous part of the entire tree.



3. Root Server

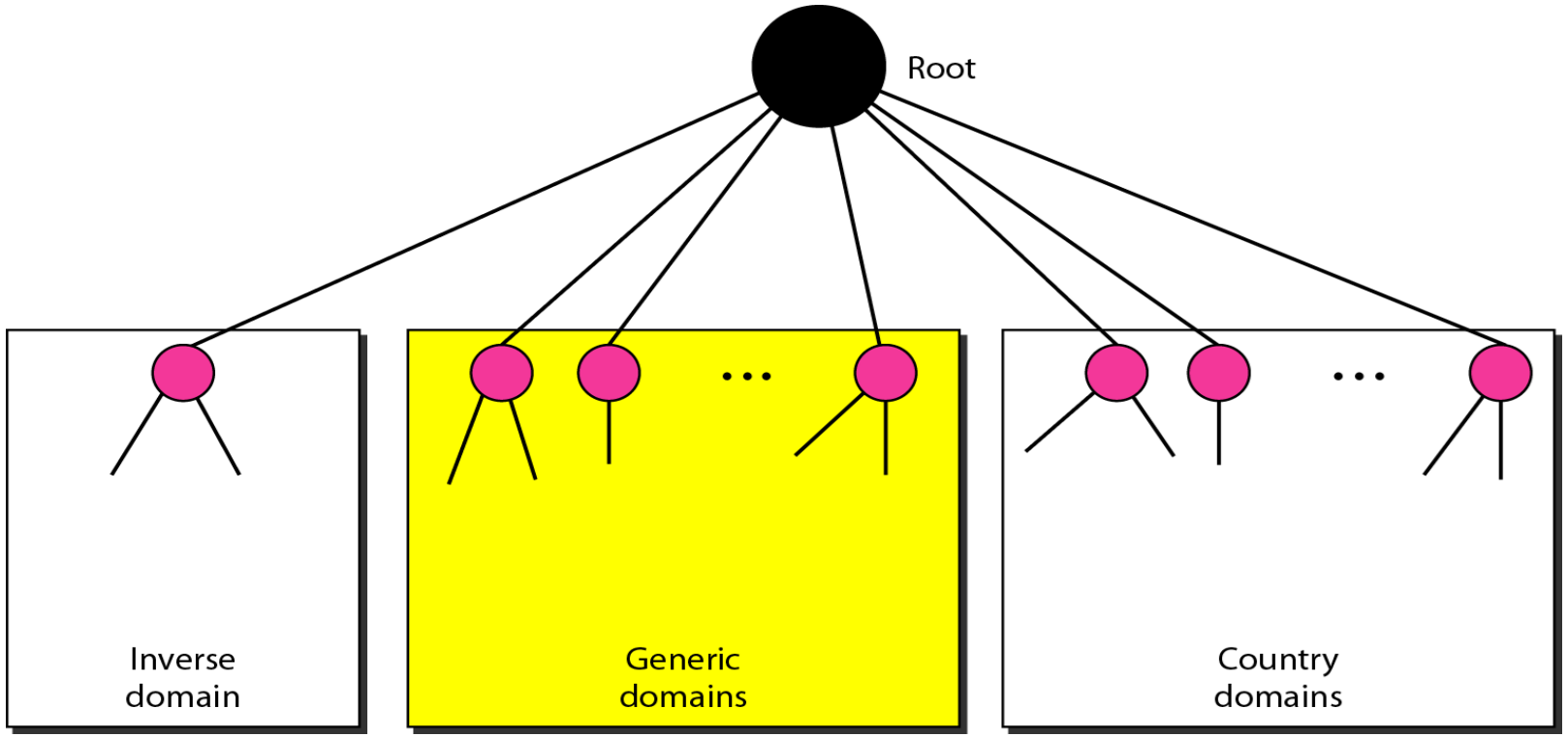
- A root server is a server whose zone consists of the whole tree. A root server usually does not store any information about domains but delegates its authority to other servers, keeping references to those servers. There are several root servers, each covering the whole domain name space. The servers are distributed all around the world.

4. Primary and Secondary Servers

- A primary server is a server that stores a file about the zone for which it is an authority. It is responsible for creating, maintaining, and updating the zone file. It stores the zone file on a local disk.
- A secondary server is a server that transfers the complete information about a zone from another server (primary or secondary) and stores the file on its local disk. The secondary server neither creates nor updates the zone files.

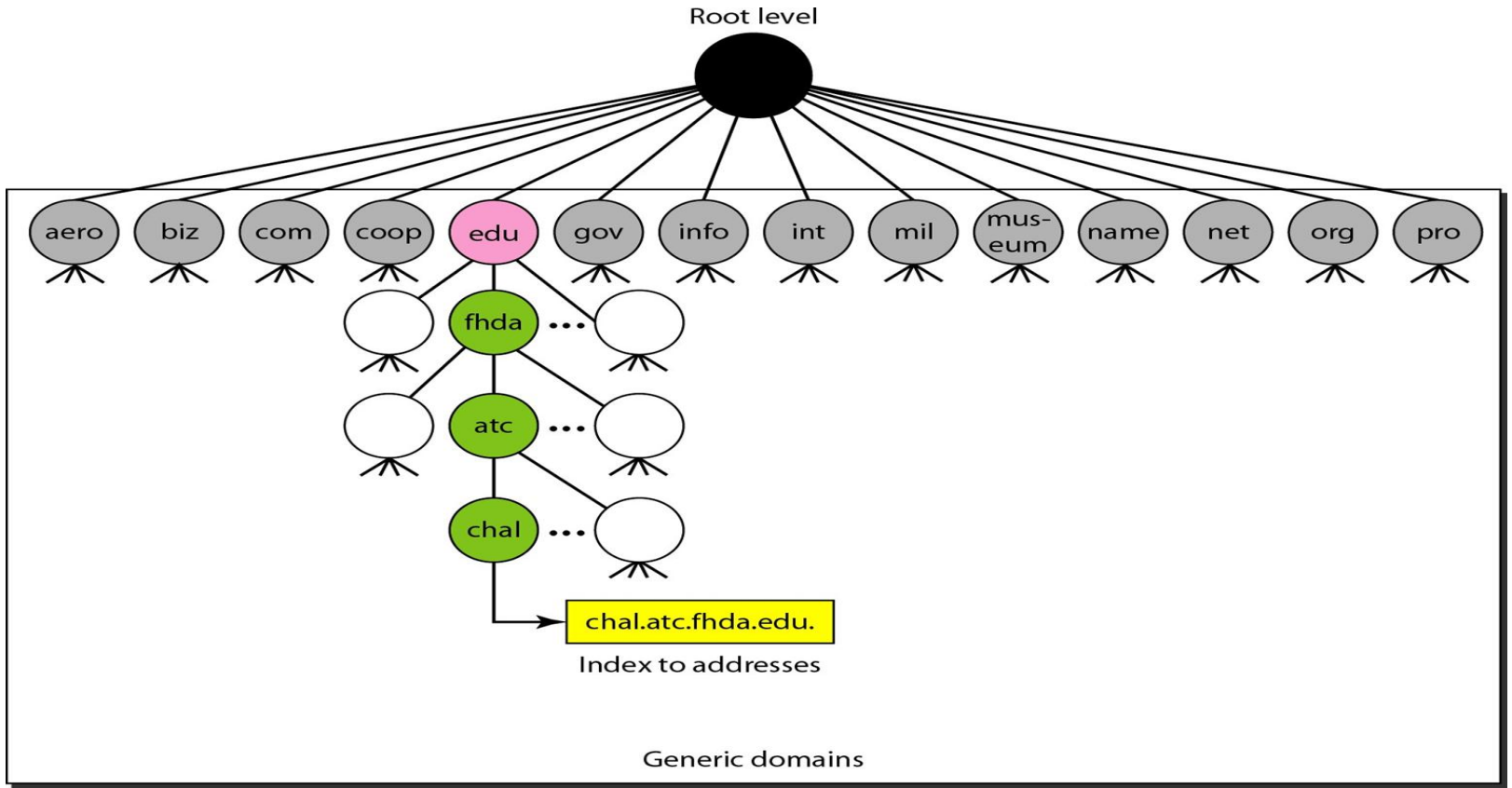
DNS IN THE INTERNET

- DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) is divided into three different sections: generic domains, country domains, and the inverse domain



1. Generic Domains

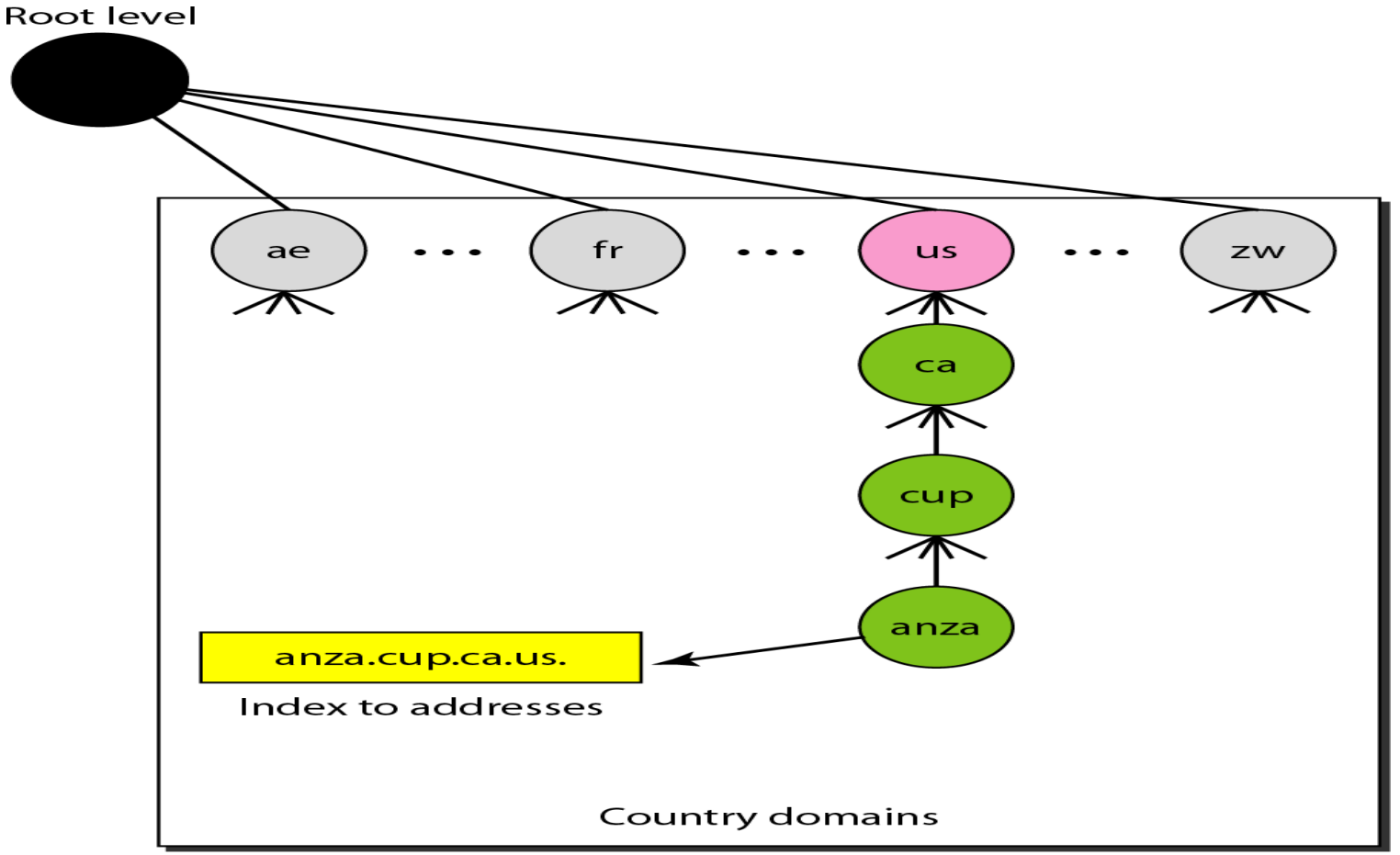
- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses three-character labels, and these labels describe the organization type.



Label	*Description*
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to "com")
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

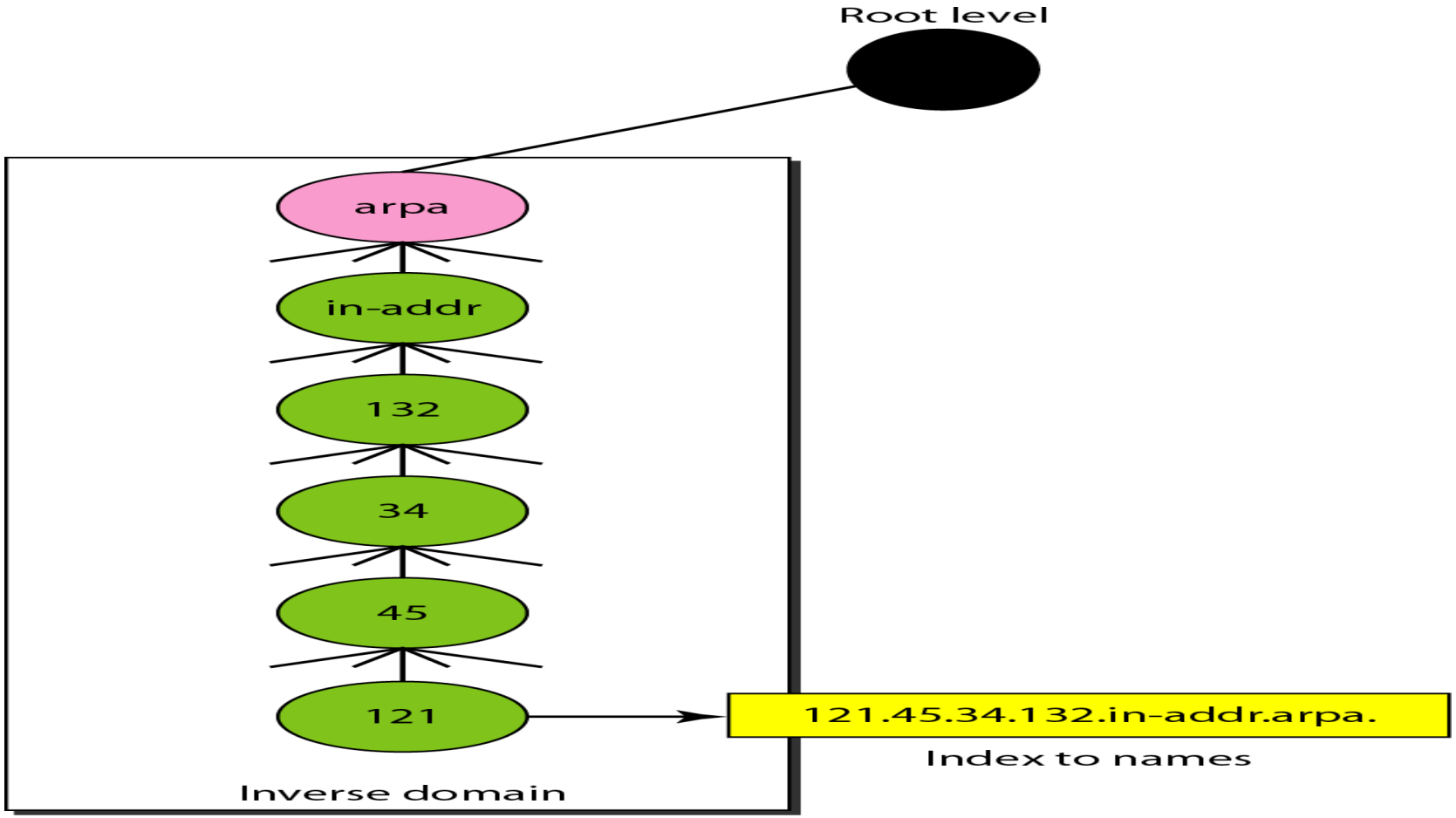
2. Country Domains

- The country domains section uses two-character country abbreviations (e.g., us for United States). Second labels can be organizational, or they can be more specific, national designations. The United States, for example, uses state abbreviations as a subdivision of us (e.g., ca.us.).



3. Inverse Domain

- The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.



RESOLUTION

✓ Mapping a name to an address or an address to a name is called name-address resolution

1. Resolver

✓ DNS is designed as a client/server application. A host that needs to map an address to a name or a name to an address calls a DNS client called a resolver. The resolver accesses the closest DNS server with a mapping request. If the server has the information, it satisfies the resolver; otherwise, it either refers the resolver to other servers or asks other servers to provide the information.

2. Mapping Names to Addresses

- In this case, the server checks the generic domains or the country domains to find the mapping.

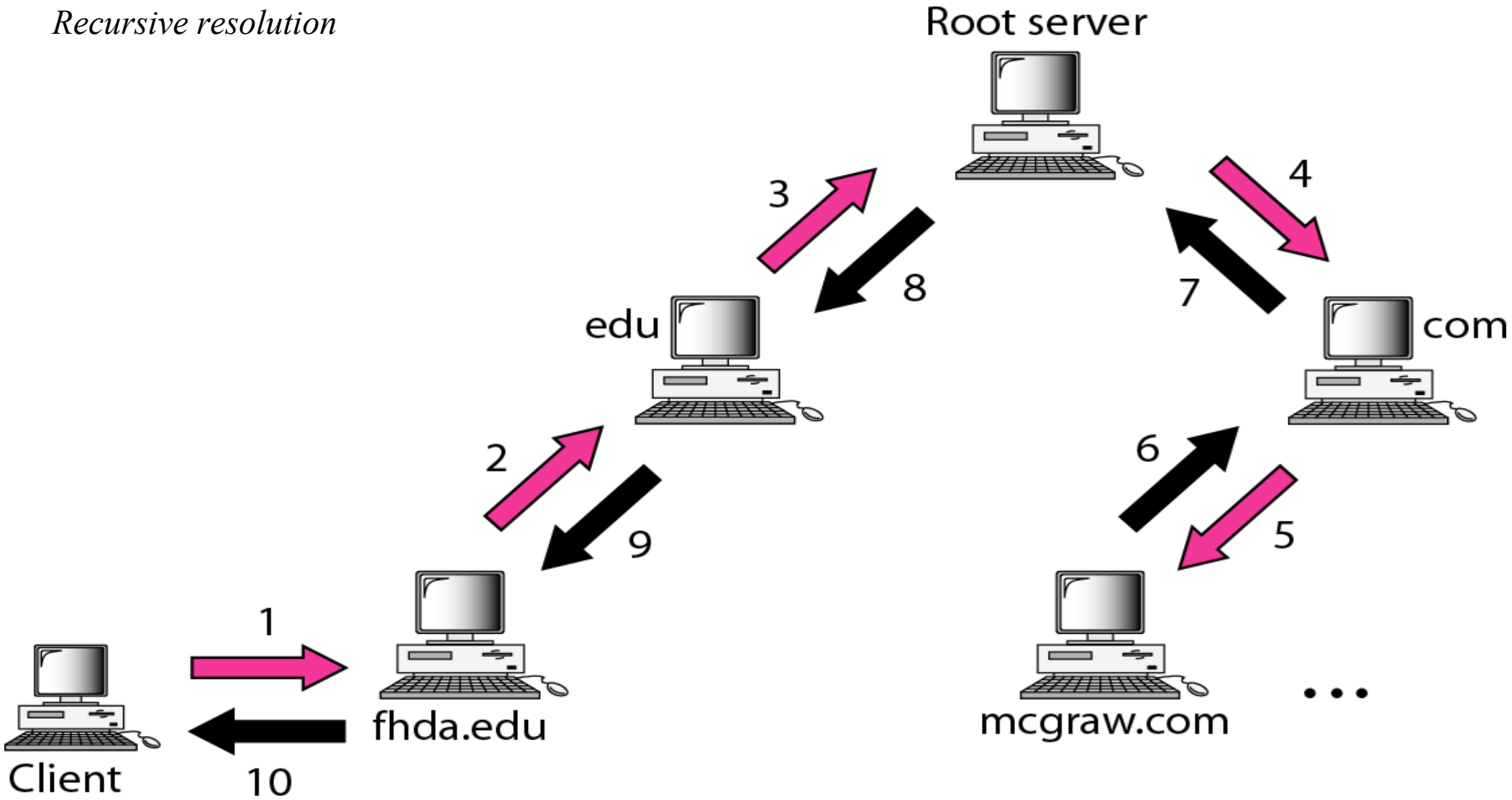
3. Mapping Addresses to Names

- To answer queries of this kind, DNS uses the inverse domain

4. Recursive Resolution

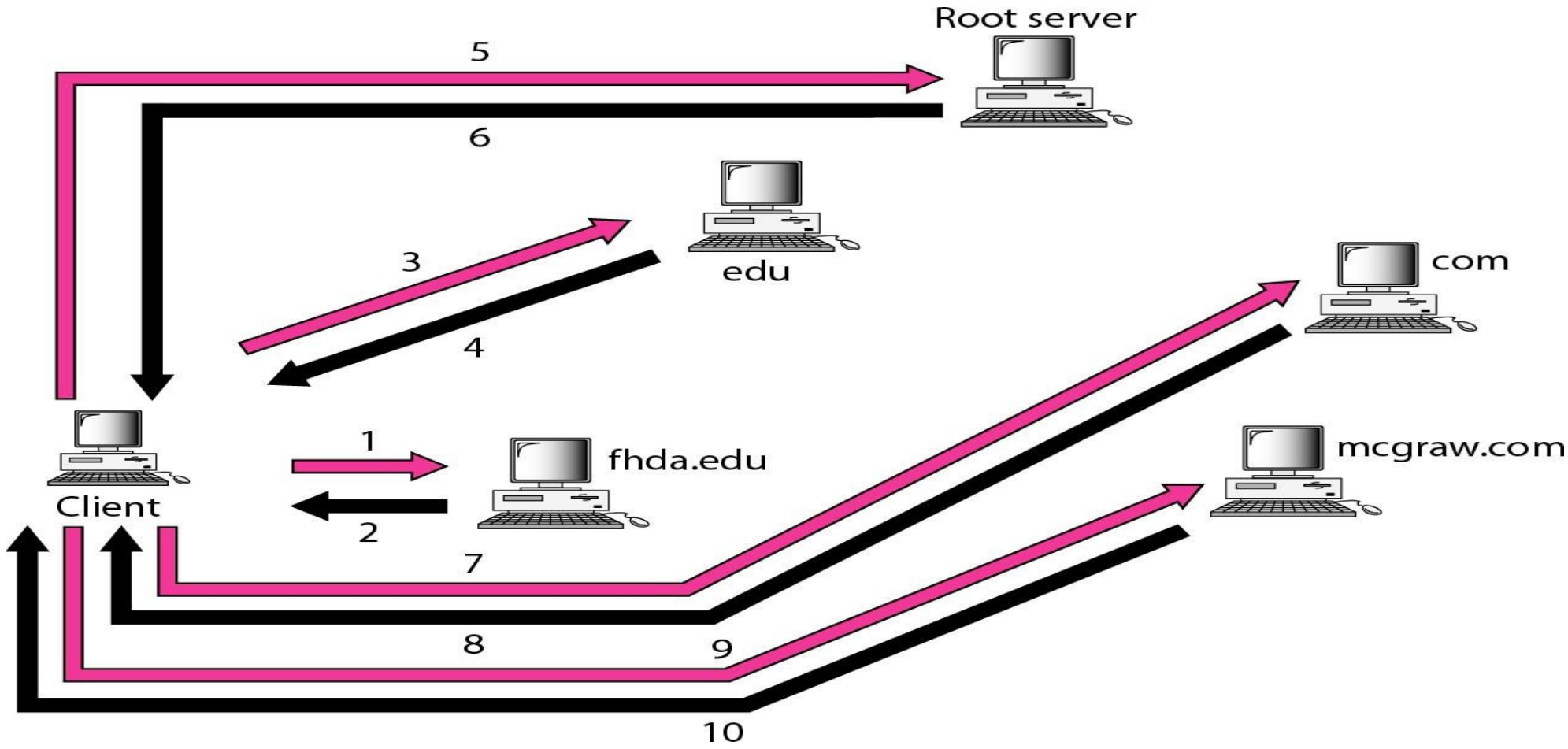
- The client (resolver) can ask for a recursive answer from a name server. This means that the resolver expects the server to supply the final answer.. When the query is finally resolved, the response travels back until it finally reaches the requesting client. This is called recursive resolution and is shown in FIG

Recursive resolution



5. Iterative Resolution

- If the client does not ask for a recursive answer, the mapping can be done iteratively. If the server is an authority for the name, it sends the answer. If it is not, it returns (to the client) the IP address of the server that it thinks can resolve the query.



6. Caching

- Each time a server receives a query for a name that is not in its domain, it needs to search its database for a server IP address. Reduction of this search time would increase efficiency. DNS handles this with a mechanism called caching.

DNS MESSAGES

- DNS has two types of messages: **query and response.** Both types have the same format.
- The query message consists of a header, and question records;
- The response message consists of a header,
    - question records, answer records, authoritative records, and additional records

TYPES OF RECORDS

- The question records are used in the question section of the query and response messages. The resource records are used in the answer, authoritative, and additional information sections of the response message.
- Question Record
- A question record is used by the client to get information from a server..
  Resource Record
- Each domain name (each node on the tree) is associated with a record called the resource record. The server database consists of resource records. Resource records are also what is returned by the server to the client.

REGISTRARS

- How are new domains added to DNS? This is done through a registrar, a commercial entity accredited by ICANN (**Internet Corporation for Assigned Names and Numbers**). A registrar first verifies that the requested domain name is unique and then enters it into the DNS database. A fee is charged.

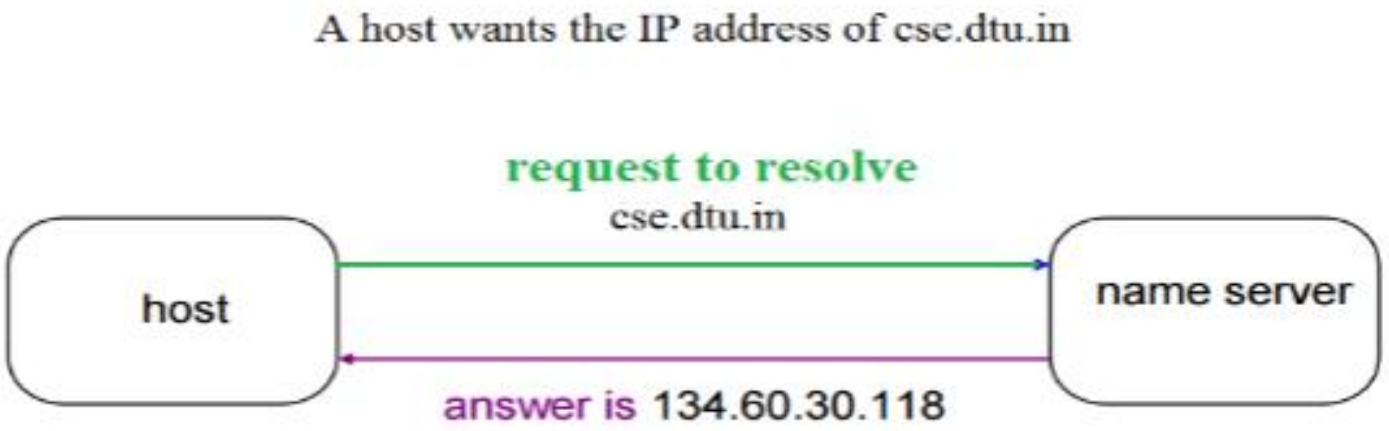
DYNAMIC DOMAIN NAME SYSTEM (DDNS)

- In DNS, when there is a change, such as adding a new host, removing a host, or changing an IP address, the change must be made to the DNS master file. The size of today's Internet does not allow for this kind of manual operation.
- The DNS master file must be updated dynamically. The Dynamic Domain Name System (DDNS) therefore was devised to respond to this need.

ENCAPSULATION

- DNS can use either UDP or TCP. In both cases the well-known port used by the server is port 53.

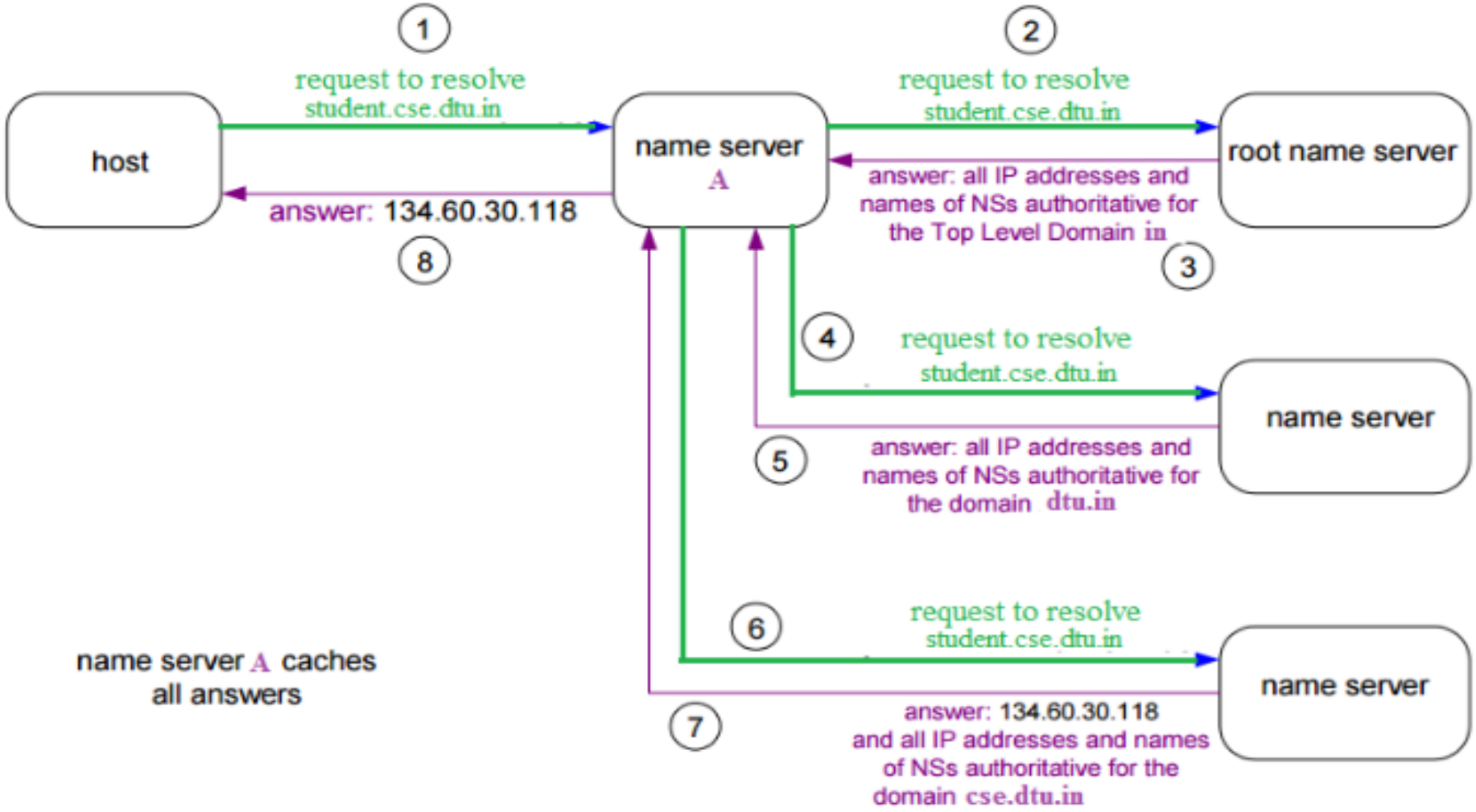
- **Mapping Name to Address Resolution:**



- The host requests the DNS name server to resolve the domain name. And the name server returns the IP address corresponding to that domain name to the host so that the host can future connect to that IP address.

- **Hierarchy of Name Servers Root name servers:** It is contacted by name servers that can not resolve the name. It contacts authoritative name server if name mapping is not known. It then gets the mapping and returns the IP address to the host.
- **Top level domain (TLD) server:** It is responsible for com, org, edu etc and all top level country domains like uk, fr, ca, in etc. They have info about authoritative domain servers and know the names and IP addresses of each authoritative name server for the second-level domains.
- **Authoritative name servers** are the organization's DNS server, providing authoritative hostName to IP mapping for organization servers. It can be maintained by an organization or service provider. In order to reach cse.dtu.in we have to ask the root DNS server, then it will point out to the top level domain server and then to authoritative domain name server which actually contains the IP address. So the authoritative domain server will return the associative ip address.

- **Domain Name Server**



- The client machine sends a request to **the local name server**, which , if root does not find the address in its database, sends a request to **the root name server** , which in turn, will route the query to an **top-level domain (TLD) or authoritative name server.**
- **The root name server** can also contain some hostName to IP address mappings. **The Top-level domain (TLD) server** always knows who the authoritative name server is. So finally the IP address is returned to the local name server which in turn returns the IP address to the host.

SNMP(Simple Network Management Protocol)

- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

For example:

- If an organization has 1000 devices then to check all devices, one by one every day, are working properly or not is a hectic task. To ease these up, Simple Network Management Protocol (SNMP) is used.
- **SNMP is an application layer protocol that uses UDP port number 161/162.**
- **SNMP is used to monitor the network, detect network faults, and sometimes even used to configure remote devices.**

SNMP components –

There are 3 components of SNMP:

1.SNMP Manager –

It is a centralized system used to monitor network. It is also known as Network Management Station (NMS)

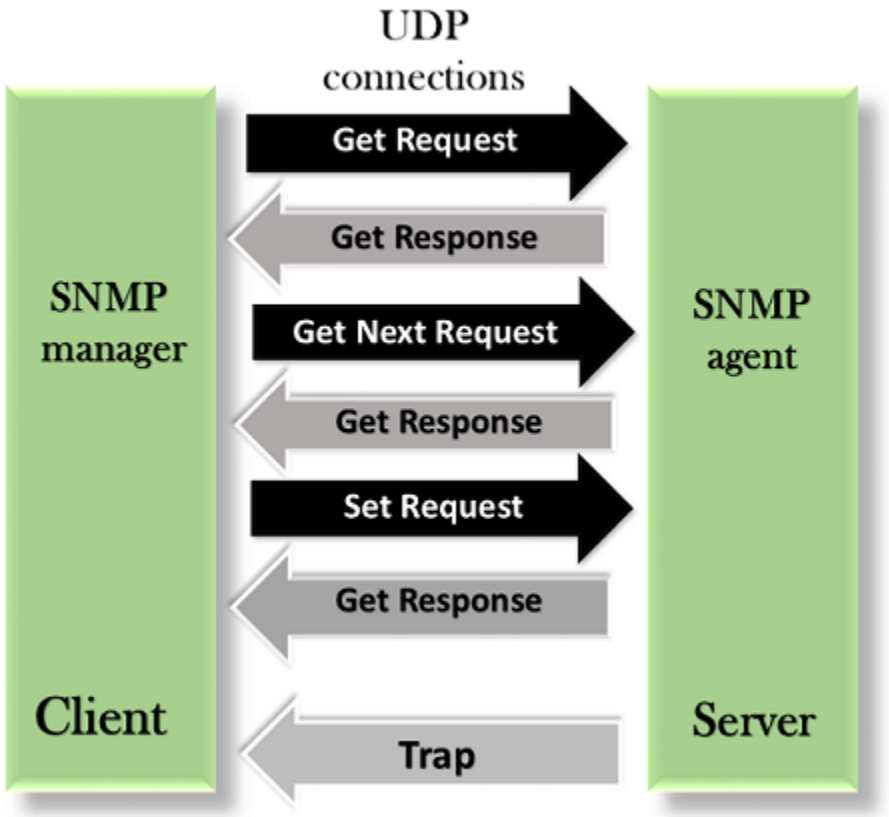
2.SNMP agent –

It is a software management software module installed on a managed device. Managed devices can be network devices like PC, routers, switches, servers, etc.

3.Management Information Base –

MIB consists of information on resources that are to be managed. This information is organized hierarchically. It consists of **objects instances** which are essentially variables.

- **SNMP defines five types of messages: GetRequest, GetNextRequest, SetRequest, GetResponse, and Trap.**



- **GetRequest:** The GetRequest message is sent from a manager (client) to the agent (server) to retrieve the value of a variable.
- **GetNextRequest:** The GetNextRequest message is sent from the manager to agent to retrieve the value of a variable. This type of message is used to retrieve the values of the entries in a table. If the manager does not know the indexes of the entries, then it will not be able to retrieve the values. In such situations, GetNextRequest message is used to define an object.
- **GetResponse:** The GetResponse message is sent from an agent to the manager in response to the GetRequest and GetNextRequest message. This message contains the value of a variable requested by the manager.
- **SetRequest:** The SetRequest message is sent from a manager to the agent to set a value in a variable.
- **Trap:** The Trap message is sent from an agent to the manager to report an event. For example, if the agent is rebooted, then it informs the manager as well as sends the time of rebooting.

SNMP security levels –

It defines the type of security algorithm performed on SNMP packets. These are used in only SNMPv3. There are 3 security levels namely:

1.noAuthNoPriv –
This (no authentication, no privacy) security level uses a community string for authentication and no encryption for privacy.

2.authNopriv – This security level (authentication, no privacy) uses HMAC with Md5 for authentication and no encryption is used for privacy.

3.authPriv – This security level (authentication, privacy) uses HMAC with Md5 or SHA for authentication and encryption uses the DES-56 algorithm.

SNMP versions –

There are 3 versions of SNMP:

1.SNMPv1 –

It uses community strings for authentication and uses UDP only.

2.SNMPv2c –

It uses community strings for authentication. It uses UDP but can be configured to use TCP.

3.SNMPv3 –

It uses Hash-based MAC with MD5 or SHA for authentication and DES-56 for privacy. This version uses TCP. Therefore, the conclusion is the higher the version of SNMP, the more secure it will be.

Electronic Mail(E-Mail)

Introduction:

- **Electronic mail (e-mail)** is a computer-based program that allows users to send and receive messages. E-mail is the electronic version of a letter, but with time and flexibility advantages. While a letter can take anywhere from a week to a couple of months to reach its intended destination, an e-mail is sent virtually almost instantly.
- Messages in the mail contain not just text but also photos, audio, and video data. A person sending an e-mail is a **sender**, and the person receiving it is the **recipient**.

What is Electronic Mail in Computer Networks?

- ✓Electronic mail is one of the most well-known network services. Electronic mail is a computer-based service that allows users to communicate with one another by exchanging messages.
- ✓Email information is transmitted via email servers and uses a variety of TCP/IP protocols. For example, the simple mail transfer protocol (SMTP) is a protocol that is used to send messages. Similarly, IMAP or POP receives messages from a mail server.

Features Of Electronic Mail

Spontaneity: In a couple of seconds, you may send a message to anybody on the globe.

Asynchronous: You may send the e-mail and let the recipient view it at their leisure.

- ✓Attachments of data, pictures, or music, frequently in compressed forms, can be delivered as an e-mail to a person anywhere in the world.
- ✓Addresses can be stored in an address book and retrieved instantly.
- ✓Through an e-mail, a user can transfer multiple copies of a message to various individuals.

But some of the limitations of this system were as follows–

- Sending a message to a group of people was inconvenient.
- Messages did not have many internet structures. So, its computer processing was complex.
- The sender never knew if the message arrived or not.
- It was not easy to hand over one's email to someone else.
- The user interface was poorly integrated.
- It was impossible to create and send messages containing a text, drawing, facsimile and voice together.

Services offered by Electronic Mail

- **Composition:** Creating messages and responses is referred to as composition.
- **Transfer:** Sending mail from the sender to the receiver is known as a transfer.
- **Reporting:** Mail delivery confirmation is known as reporting. It allows users to see if
their mail has been delivered, misplaced, or rejected.
- **Displaying:** It refers to presenting messages so that the user can understand them.
- **Disposition:** This stage concerns the recipient's actions after receiving mail, such as saving it, deleting it before reading it, or after reading it.

Components Of Electronic Mail

- The following are the essential components of an e-mail system:

1.User Agent (UA)

2.Message Transfer Agent (MTA)

3.Message Access Agent

1. User Agent (UA)

➢The User-Agent is a simple software that sends and receives mail. It is also known as a mail reader. It supports a wide range of instructions for sending, receiving, and replying to messages and manipulating mailboxes.

➢Some of the services supplied by the User-Agent are listed below:

- Reading a Message
- Sending a reply to a Message
- Message Composition
- Forwarding a Message
- Handling the Message

2. Message Transfer Agent

- The Message Transfer Agent manages the actual e-mail transfer operation (MTA). Simple Mail Transfer Protocol sends messages from one MTA to another. A system must have a client MTA and a system MTA to send an e-mail. If the recipients are connected to the same computer, it sends mail to their mailboxes. If the destination mailbox is on another computer, it sends mail to the receiver's MTA.

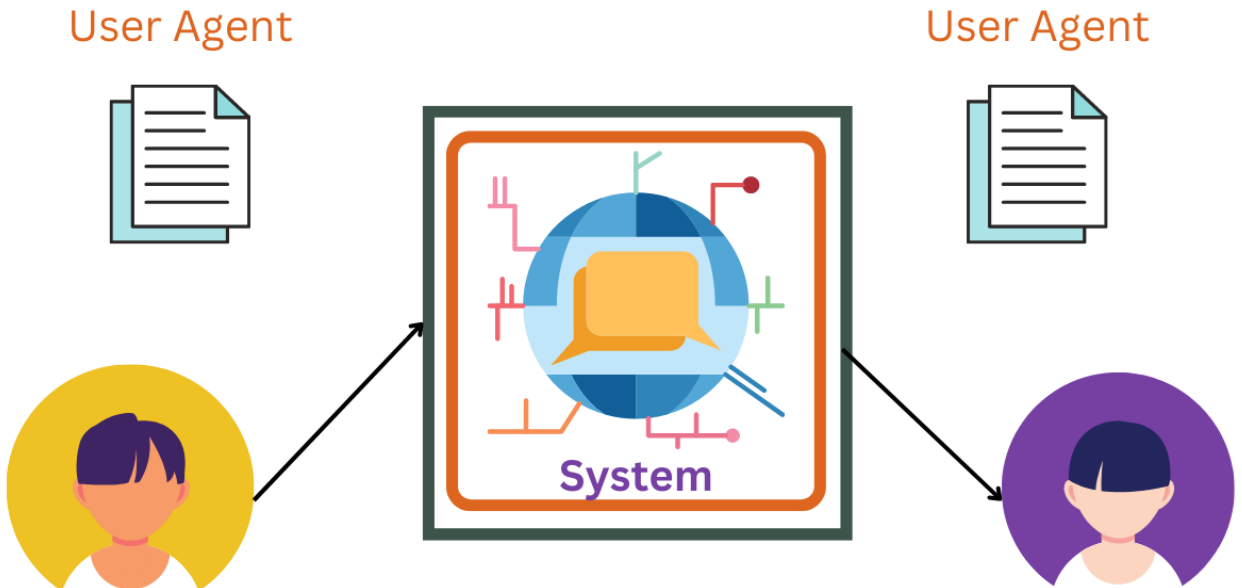
3. Message Access Agent

- The Simple Mail Transfer Protocol is used for the first and second stages of e-mail delivery.
- The pull protocol is mainly required at the third stage of e-mail delivery, and the message access agent is used at this point.
- POP and IMAP4 are the two protocols used to access messages.

Architecture of Electronic Mail

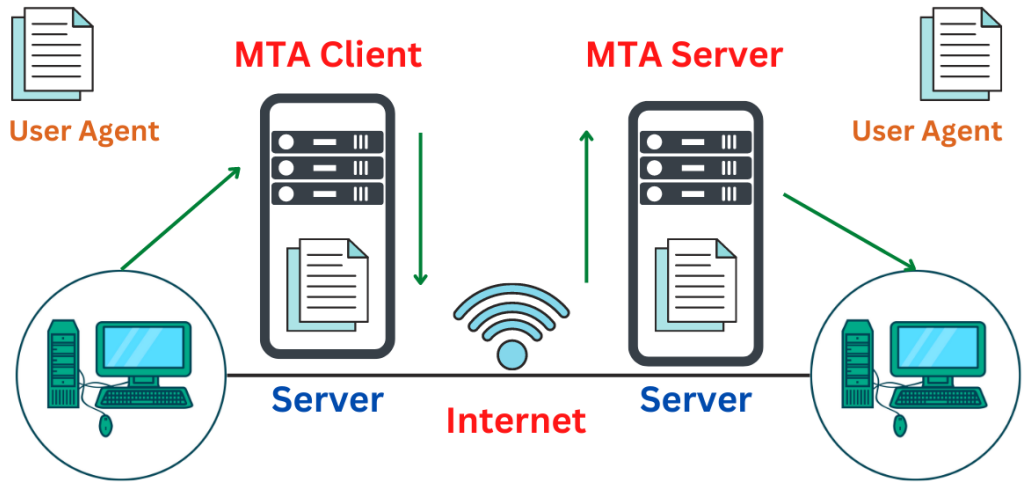
First Scenario

- In the first scenario, two user agents are required. The sender and recipient of the e-mail share the same machine directly connected to the server.
- For example, let us consider two user agents, Ninja1 and Ninja2. When Ninja1 sends an e-mail to Ninja2, the user agent (UA) programme is used to prepare the message. Following that, this e-mail gets saved in the Ninja2 inbox.



Second Scenario

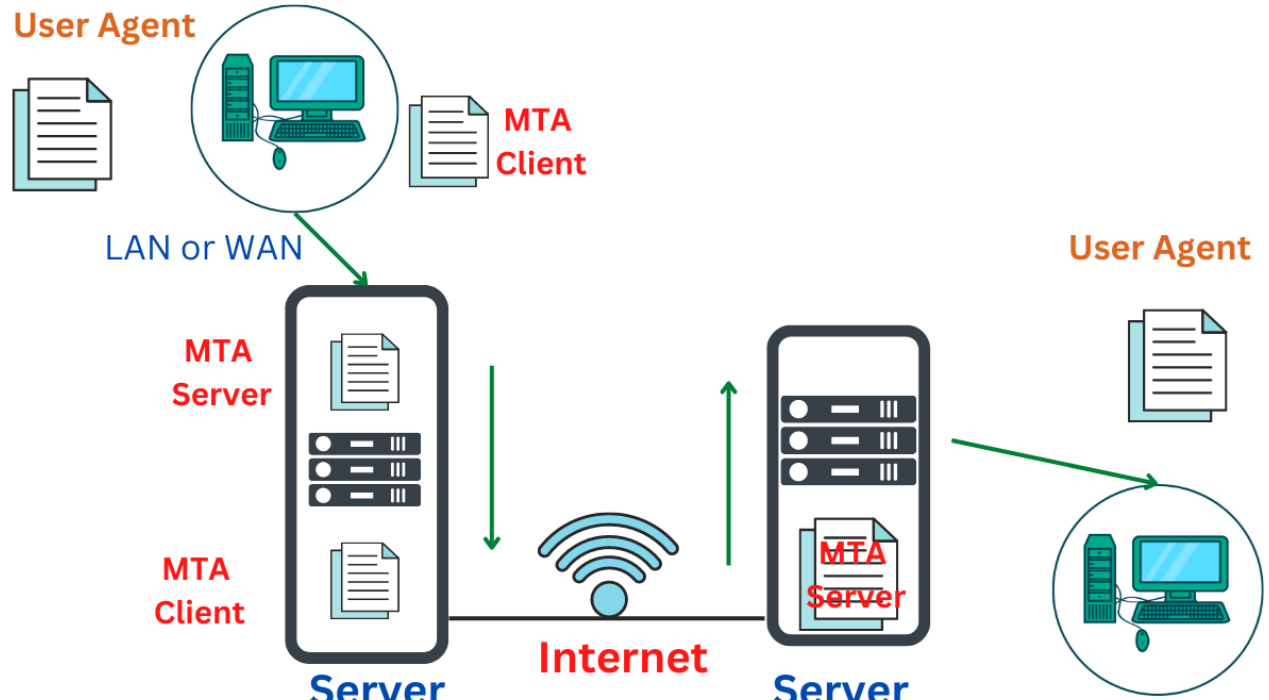
- In this case, the sender and recipient of an e-mail are essentially users on two different machines over the internet. User-Agents and Message Transfer Agents(MTA) are required in this scenario.



Take, for example, two user agents (Ninja1 and Ninja2), as illustrated in the diagram. When Ninja1 sends an e-mail to Ninja2, the user agent (UA) and message transfer agents (MTAs) programmes prepare the e-mail for transmission over the internet. Following that, this e-mail gets stored in Ninja2's inbox.

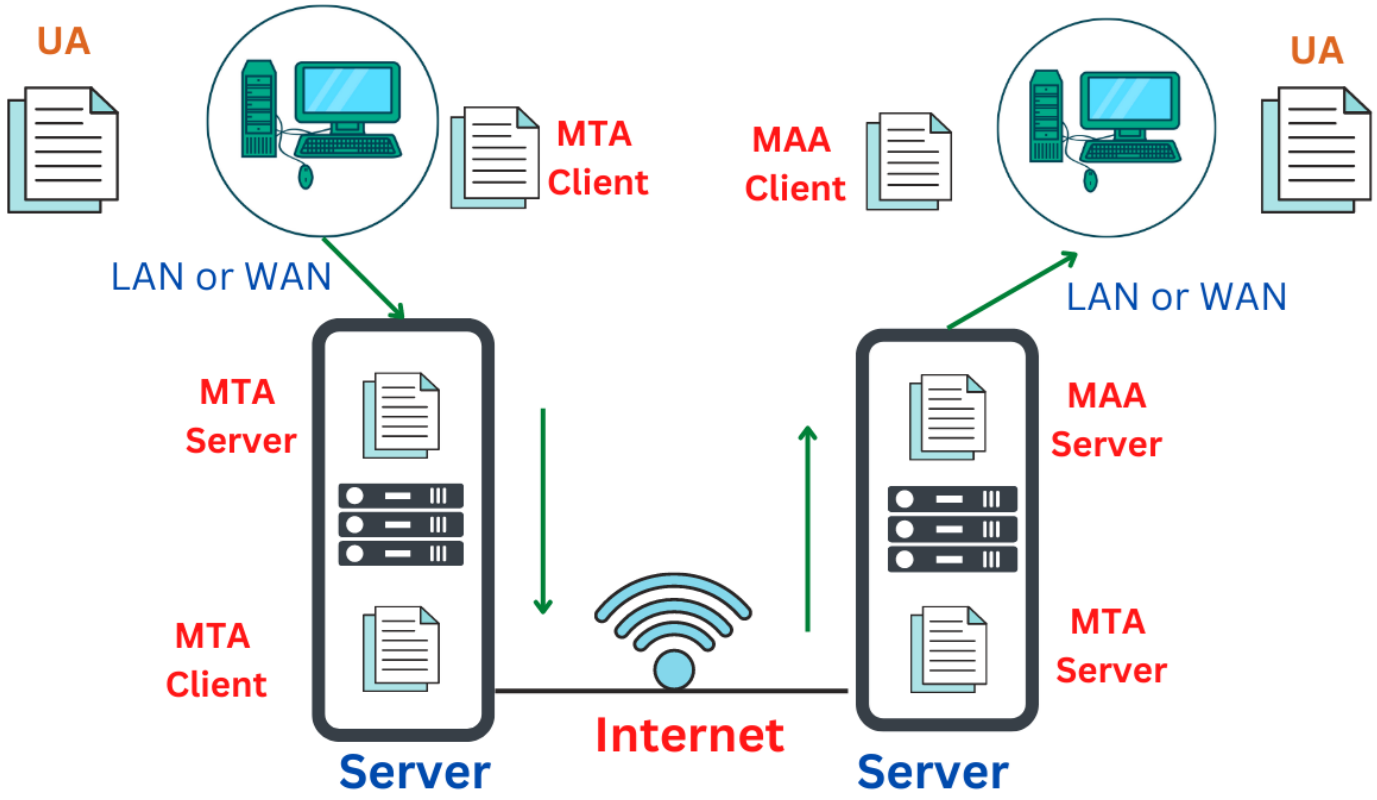
Third Scenario

- The sender is connected to the system by a point-to-point WAN, which can be a dial-up modem or a cable modem in this case. On the other hand, the receiver is directly attached to the system, as it was in the second scenario.
- The sender also needs a User agent (UA) to prepare the message in this situation. After preparing the statement, the sender delivers it over LAN or WAN via a pair of MTAs.



Fourth Scenario

- In this scenario, the recipient is linked to the mail server via WAN or LAN. When the message arrives, the recipient must retrieve it, which needs additional client/server agents. This scenario requires two user agents (UAs), two pairs of message transfer agents (MTAs), and a couple of message access agents (MAAs).



Simple Mail Transfer Protocol(SMTP)

- Email is emerging as one of the most valuable services on the internet today. Most internet systems **use SMTP as a method to transfer mail from one user to another. SMTP is a push protocol and is used to send the mail** whereas **POP (post office protocol) or IMAP (internet message access protocol) are used to retrieve those emails at the receiver's side**.
- SMTP is an application layer protocol. The client who wants to send the mail opens a TCP connection to the SMTP server and then sends the mail across the connection.
- The SMTP server is an always-on listening mode. As soon as it listens for a TCP connection from any client, the SMTP process initiates a connection through port 25.
- After successfully establishing a TCP connection the client process sends the mail instantly.

Components of SMTP:

- **The user agent (UA) : it** prepares the message, creates the envelope and then puts the message in the envelope.
- **The mail transfer agent (MTA) : it** transfers this mail across the internet.

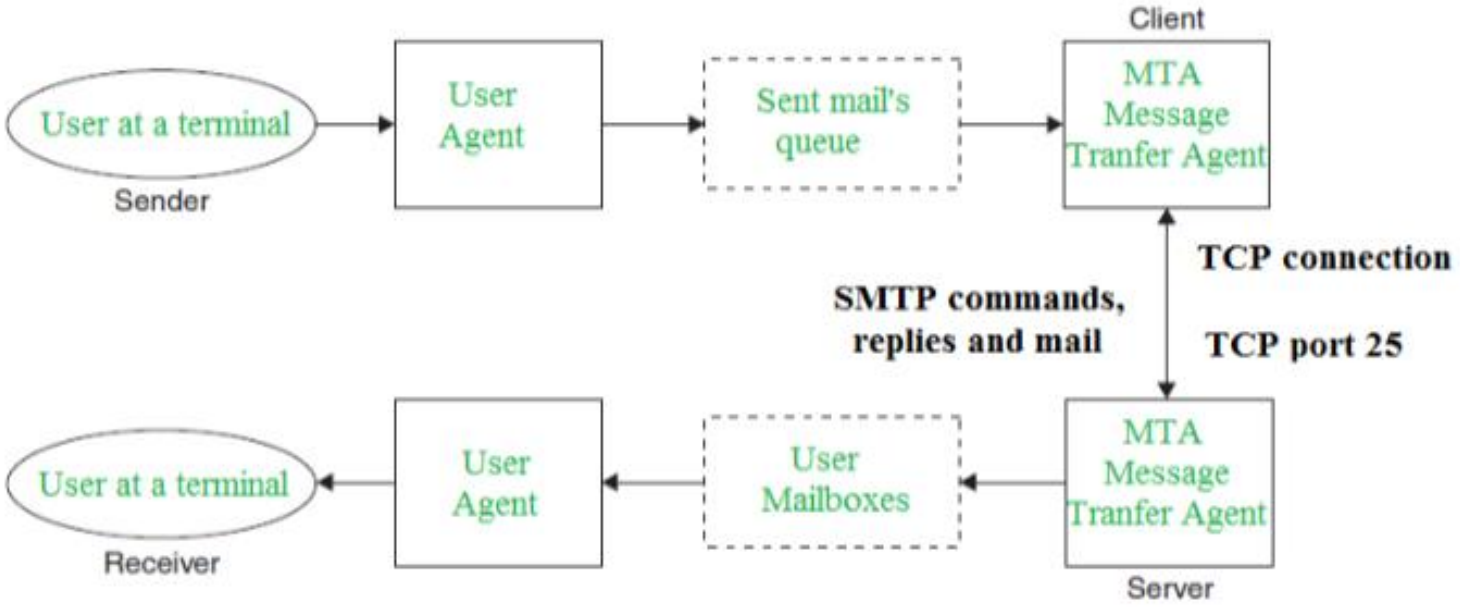
SMTP Protocol Method:

1. **Store-and-Forward Method** The store and forward method is used within an organization.
2. **End-to-End Method** Mainly the end-to-end method is used to communicate between the different organizations.

- An SMTP client who wants to send the mail will contact the destination's host SMTP directly, in order to send the mail to the destination.
- The SMTP server will keep the mail to itself until it is successfully copied to the receiver's SMTP.
- The client SMTP is the one that initiates the session so let us call it client- SMTP and the server SMTP is the one that responds to the session request so let us call it receiver-SMTP.
- The client- SMTP will start the session and the receiver-SMTP will respond to the request.

Architecture of SMTP

- In the SMTP model user deals with the user agent (UA), for example, Microsoft Outlook, Netscape, Mozilla, etc. In order to exchange the mail using TCP, MTA is used.
- The user sending the mail doesn't have to deal with MTA as it is the responsibility of the system admin to set up a local MTA. The MTA maintains a small queue of mails so that it can schedule repeat delivery of mails in case the receiver is not available. The MTA delivers the mail to the mailboxes and the information can later be downloaded by the user agents.
-



- **Communication between sender and the receiver :**
- The sender's user agent prepares the message and sends it to the MTA. The MTA's responsibility is to transfer the mail across the network to the receiver's MTA. To send mails, a system must have a client MTA, and to receive mails, a system must have a server MTA.
- **SENDING EMAIL:**
  Mail is sent by a series of request and response messages between the client and the server. The message which is sent across consists of a header and a body. A null line is used to terminate the mail header and everything after the null line is considered as the body of the message, which is a sequence of ASCII characters. The message body contains the actual information read by the receipt.

- **RECEIVING EMAIL:**
  The user agent at the server-side checks the mailboxes at a particular time of intervals. If any information is received, it informs the user about the mail. When the user tries to read the mail it displays a list of emails with a short description of each mail in the mailbox. By selecting any of the mail users can view its contents on the terminal.

Some SMTP Commands:

- HELO – Identifies the client to the server, fully qualified domain name, only sent once per session
- MAIL – Initiate a message transfer, fully qualified domain of originator
- RCPT – Follows MAIL, identifies an addressee, typically the fully qualified name of the addressee, and for multiple addressees use one RCPT for each addressee
- DATA – send data line by line

Advantages of SMTP:

- If necessary, the users can have a dedicated server.
- It allows for bulk mailing.
- Low cost and wide coverage area.
- Offer choices for email tracking.
- reliable and prompt email delivery.

Disadvantages of SMTP:

- SMTP's common port can be blocked by several firewalls.
- SMTP security is a bigger problem.
- Its simplicity restricts how useful it can be.
- Just 7 bit ASCII characters can be used.
- If a message is longer than a certain length, SMTP servers may reject the entire message.
- Delivering your message will typically involve additional back-and-forth processing between servers, which will delay sending and raise the likelihood that it won't be sent.

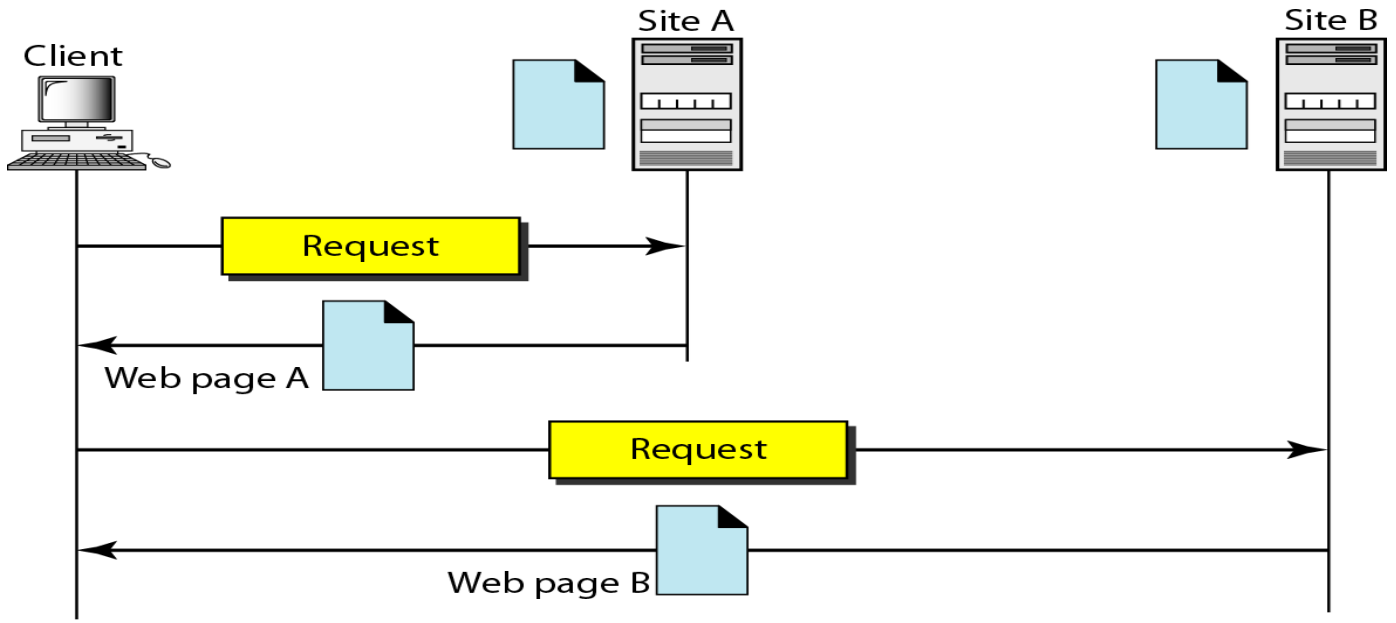
World Wide Web (WWW)

- The **World Wide Web** is abbreviated as WWW and is commonly known as the web. The WWW was initiated by CERN (European library for Nuclear Research) in 1989.
- **WWW can be defined as the collection of different websites around the world, containing different information shared via local servers(or computers).**

History:

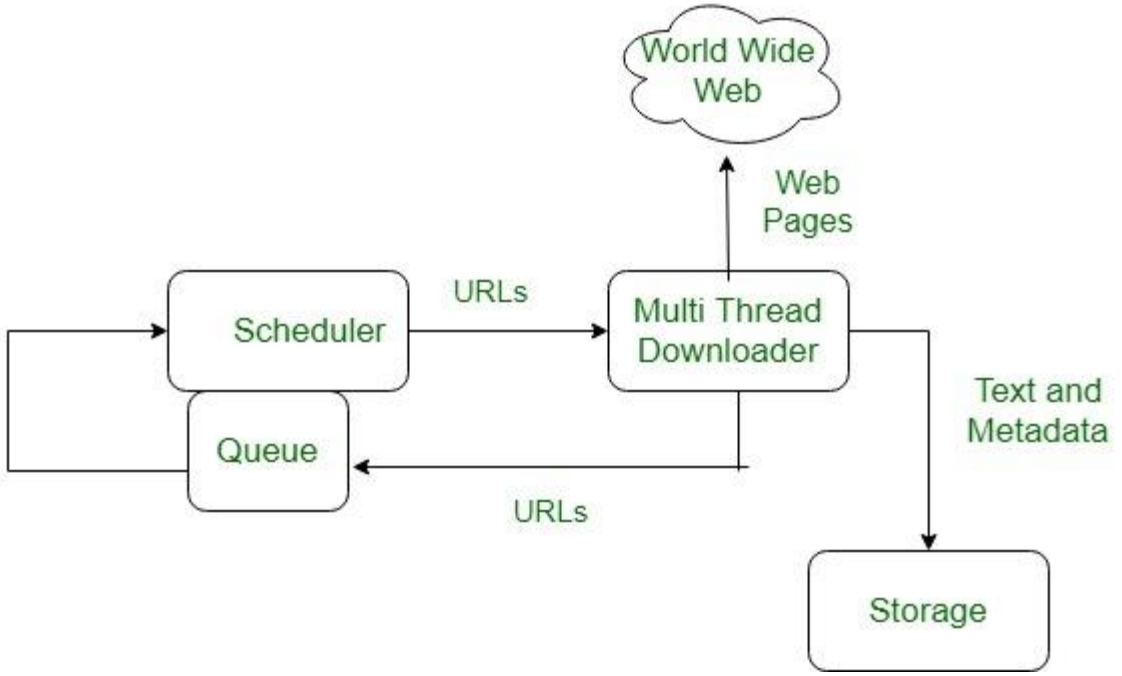
It is a project created, by Timothy Berner Lee in 1989, for researchers to work together effectively at CERN. is an organization, named the World Wide Web Consortium (W3C), which was developed for further development of the web. This organization is directed by Tim Berner's Lee, aka the father of the web.

- The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites as shown in fig.



System Architecture:

- **From the user's point of view**, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc are the popular ones.
- **The browser** fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.
- The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

Working of WWW:

- **The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).**
- A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet.
- Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal.
- Web browsers can be used for several tasks including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

Components of WWW

Client (Browser):

- Web browser is a program, which is used to communicate with web server on the Internet.
- Each browser consists of three parts: a controller, client protocol and interpreter.
- The controller receives input from input device and use the programs to access the documents.
- After accessing the document, the controller uses one of the interpreters to display the document on the screen.

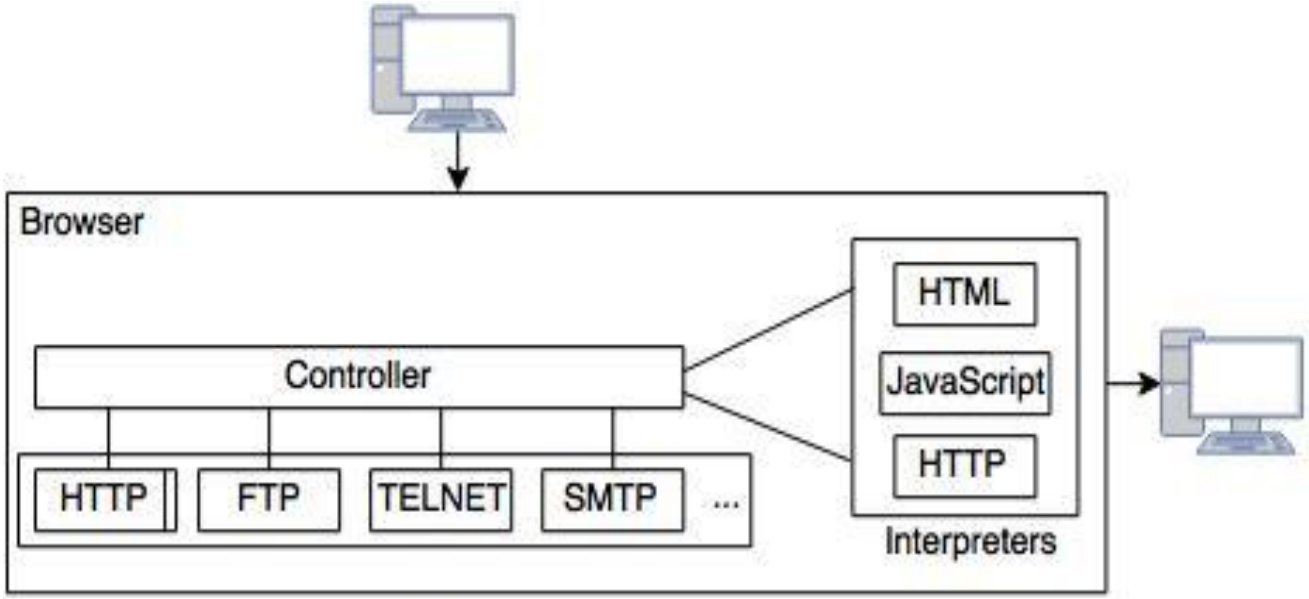


Fig: CIlient (Browser)

Server:

- A computer which is available for the network resources and provides service to the other computer on request is known as server.
- The web pages are stored at the server.
- Server accepts a TCP connection from a client browser.
- It gets the name of the file required.
- Server gets the stored file. Returns the file to the client and releases the top connection.

Uniform Resource Locater (URL):

- The URL is a standard for specifying any kind of information on the Internet.
- The URL consists of four parts: protocol, host computer, port and path.
- The protocol is the client or server program which is used to retrieve the document or file. The protocol can be ftp or http.
- The host is the name of computer on which the information is located.
- The URL can optionally contain the port number and it is separated from the host name by a colon.
- Path is the pathname of the file where the file is stored.

HTML:

- HTML is an abbreviation of Hypertext Markup Language.
- It is generally used for creating web pages.
- It is mainly used to define the contents, structure, and organization of the web page.

HyperText Transfer Protocol (HTTP):

- it specifies communication of browser and server.

Advantages of WWW

- Given below are the benefits offered by WWW:
- It mainly provides all the information for Free.
- Provides rapid Interactive way of Communication.
- It is accessible from anywhere.
- It has become the Global source of media.
- It mainly facilitates the exchange of a huge volume of data.

➢ **The documents in the WWW can be grouped into three broad categories: static, dynamic, and active. The category is based on the time at which the contents of the document are determined.**

Static Documents:

- Static documents are fixed-content documents that are created and stored in a server. The client can get only a copy of the document. When a client accesses the document, a copy of the document is sent. The user can then use a browsing program to display the document.
- ***HTML***
- Hypertext Markup Language (HTML) is a language for creating Web pages.

Dynamic Documents

- A dynamic document is created by a Web server whenever a browser requests the document. When a request arrives, the Web server runs an application program or a script that creates the dynamic document. The server returns the output of the program or script as a response to the browser that requested the document.
- A very simple example of a dynamic document is the retrieval of the time and date from a server. Time and date are kinds of information that are dynamic inthatthey changefrommoment to moment. The client can ask the server to run a program such as the *date program in UNIX and send the result of the program to* the client.

Common Gateway Interface (CGI)

- The Common Gateway Interface (CGI) is a technologythat creates and handles dynamic documents.
- Hypertext Preprocessor (pHP), which uses the Perl language; Java Server Pages (JSP), which uses the Java language for scripting; Active Server Pages (ASP), a Microsoft product which uses Visual Basic language for scripting; and ColdFusion, which embeds SQL database queries in the HTML document.
- Dynamic documents are sometimes referred to as server-site dynamic documents.

Active Documents

- For many applications, we need a program or a script to be run at the client site. These are called active documents.

HTTP(HyperText Transfer Protocol)

- HTTP stands for **HyperText Transfer Protocol**.
- It is a protocol used to access the data on the World Wide Web (www).
- The HTTP protocol can be used to transfer the data in the form of plain text, hypertext, audio, video, and so on.
- This protocol is known as HyperText Transfer Protocol because of its efficiency that allows us to use in a hypertext environment where there are rapid jumps from one document to another document.
- HTTP is similar to the FTP as it also transfers the files from one host to another host. But, HTTP is simpler than FTP as HTTP uses only one connection, i.e., no control connection to transfer the files.
- HTTP is used to carry the data in the form of MIME-like format.

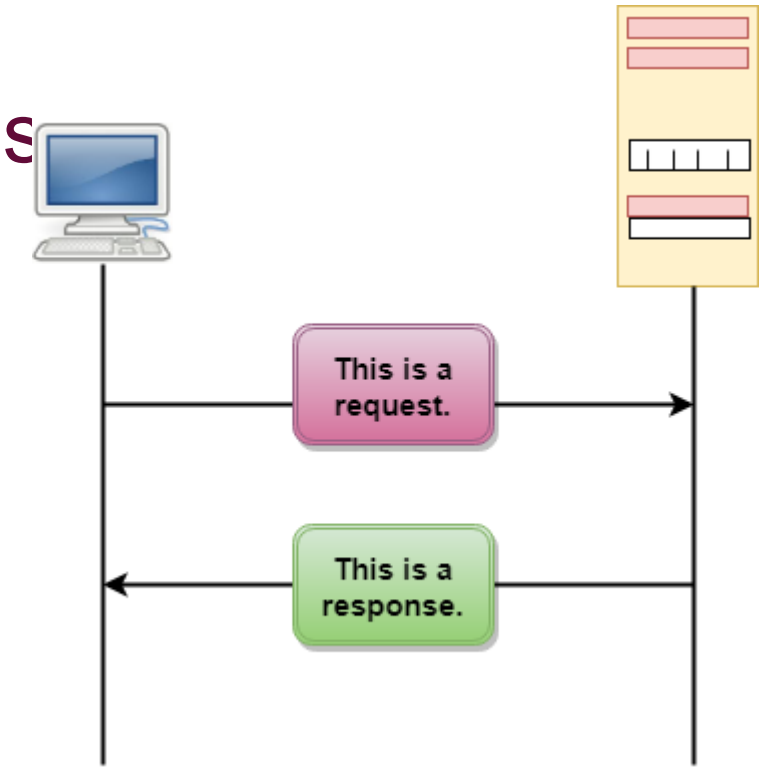
- HTTP is similar to SMTP as the data is transferred between client and server. The HTTP differs from the SMTP in the way the messages are sent from the client to the server and from server to the client. SMTP messages are stored and forwarded while HTTP messages are delivered immediately.

Features of HTTP:

- **Connectionless protocol:** HTTP is a connectionless protocol. HTTP client initiates a request and waits for a response from the server. When the server receives the request, the server processes the request and sends back the response to the HTTP client after which the client disconnects the connection. The connection between client and server exist only during the current request and response time only.
- **Media independent:** HTTP protocol is a media independent as data can be sent as long as both the client and server know how to handle the data content. It is required for both the client and server to specify the content type in MIME-type header.

Stateless: HTTP is a stateless protocol as both the client and server know each other only during the current request. Due to this nature of the protocol, both the client and server do not retain the information between various requests of the web pages.

HTTP Transactions



The above figure shows the HTTP transaction between client and server. The client initiates a transaction by sending a request message to the server. The server replies to the request message by sending a response message.

- **HTTP messages are of two types: request and response. Both the message types follow the same message format.**

  Request Message: The request message is sent by the client that consists of a request line, headers, and sometimes a body.

  Response Message: The response message is sent by the server to the client that consists of a status line, headers, and sometimes a body.

Uniform Resource Locator (URL)

- A client that wants to access the document in an internet needs an address and to facilitate the access of documents, the HTTP uses the concept of Uniform Resource Locator (URL).
- The Uniform Resource Locator (URL) is a standard way of specifying any kind of information on the internet.
- The URL defines four parts: method, host computer, port, and path.



- **Method:** The method is the protocol used to retrieve the document from a server. For example, HTTP.
- **Host:** The host is the computer where the information is stored, and the computer is given an alias name. Web pages are mainly stored in the computers and the computers are given an alias name that begins with the characters "www". This field is not mandatory.
- **Port:** The URL can also contain the port number of the server, but it's an optional field. If the port number is included, then it must come between the host and path and it should be separated from the host by a colon.
- **Path:** Path is the pathname of the file where the information is stored. The path itself contain slashes that separate the directories from the subdirectories and files.

- **How it works ?**
  First of all, whenever we want to open any website then first we open web browser after that we will type URL of that website (e.g., www.facebook.com). This URL is now sent to Domain Name Server (DNS). Then DNS first check records for this URL in their database, then DNS will return IP address to web browser corresponding to this URL. Now browser is able to sent request to actual server.
- After server sends data to client, connection will be closed. If we want something else from server we should have to re-establish connection between client and server.

Characteristics of HTTP: HTTP is IP based communication protocol which is used to deliver data from server to client or vice-versa.

1.Server processes a request, which is raised by client and also server and client knows each other only during current request and response period.

2.Any type of content can be exchanged as long as server and client are compatible with it.

3.Once data is exchanged then servers and client are no more connected with each other.

4.It is a request and response protocol based on client and server requirements.

5.It is connection less protocol because after connection is closed, server does not remember anything about client and client does not remember anything about server.

6.It is stateless protocol because both client and server does not expecting anything from each other but they are still able to communicate.

Advantages :

- Memory usage and CPU usage are low because of less simultaneous connections.
- Since there are few TCP connections hence network congestion are less.
- Since handshaking is done at initial connection stage, then latency is reduced because there is no further need of handshaking for subsequent requests.
- The error can be reports without closing connection.
- HTTP allows HTTP pipe-lining of request or response.

Disadvantages :

- HTTP requires high power to establish communication and transfer data.
- HTTP is less secure, because it does not uses any encryption method like https use TLS to encrypt normal http requests and response.
- HTTP does not offer genuine exchange of data because it is less secure.
- Client does not close connection until it receives complete data from server and hence server needs to wait for data completion and cannot be available for other clients during this time.

Streaming of Audio and Video